

DATA VISUALIZATION FINAL PRESENTATION

American Food Insecurity

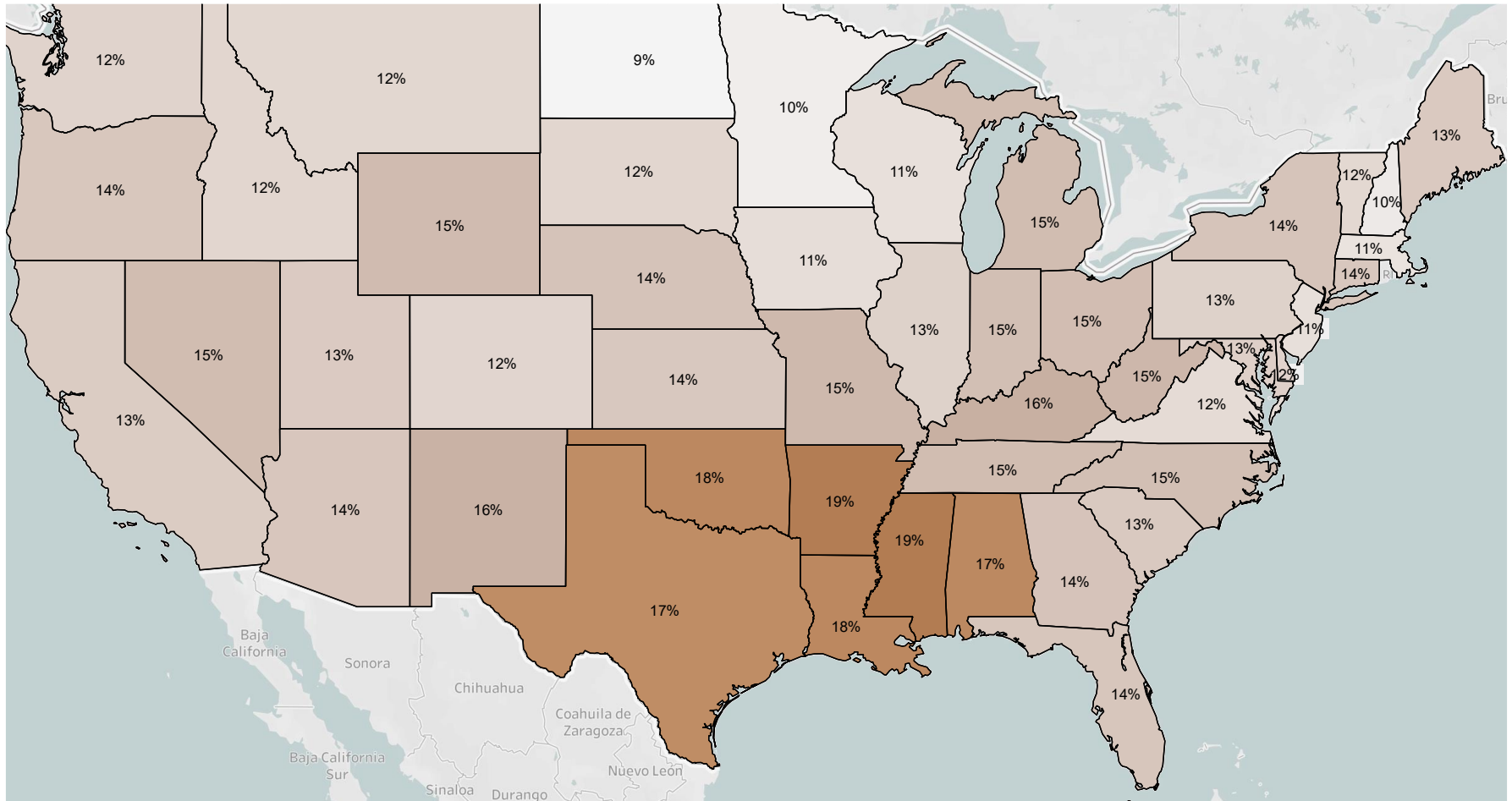
No Way Out

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Millions of Americans Don't Have Enough Food

Despite a \$32 trillion GDP, food inequality in America remains deeply entrenched. Over 45 million people, across 18 million households, struggled to afford enough food in 2024 — this included 14.1 million children. The burden falls hardest on the South, where rates are nearly double those in the lowest-insecurity states.

American food inequality by state



Research Framework

Aims

- Investigate why food insecurity persists in the wealthiest nation on earth — reframing it as a problem of distribution, not scarcity.
- Trace the chain from poverty to food insecurity to health outcomes across U.S. counties.
- Identify where intervention should be prioritized — the counties where all three crises compound.

Audience

- Policymakers and public health officials with authority to allocate budget and resources to underserved communities.
- Researchers studying food security, poverty, and structural inequality.
- Built for stakeholders who need to act on where, not just understand why.

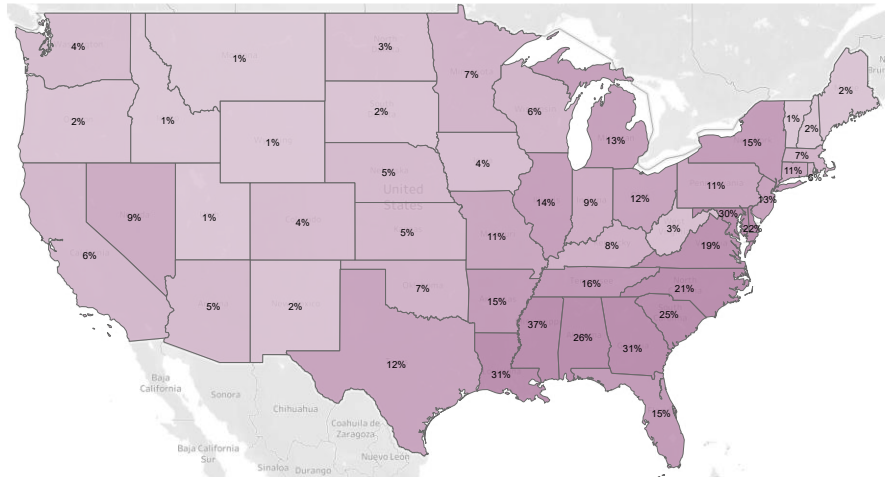
Methodology

- County- and city-level analysis joining CDC 500 Cities (2019), USDA Food Access Research Atlas, Feeding America's Map the Meal Gap (2017), and U.S. Census data.
- Cleaning, merging, and derived metrics (budget shortfalls, z-scored rankings) built in Python / pandas.
- Visualized in Tableau and polished in Inkscape for standalone clarity.

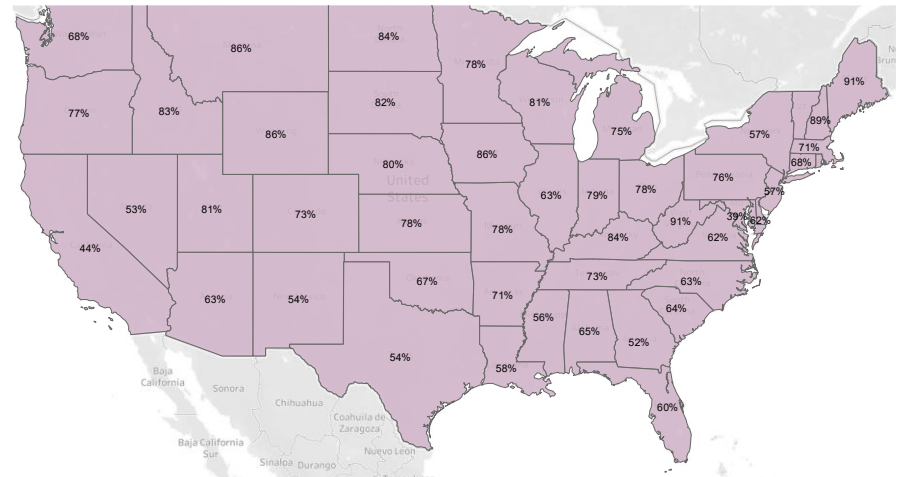
The Problem is Driven by an Ethnic Divide

Black and Hispanic households face food insecurity at rates nearly double that of White households. In the South, where these communities exceed 30% of the population, this disparity compounds into a regional crisis.

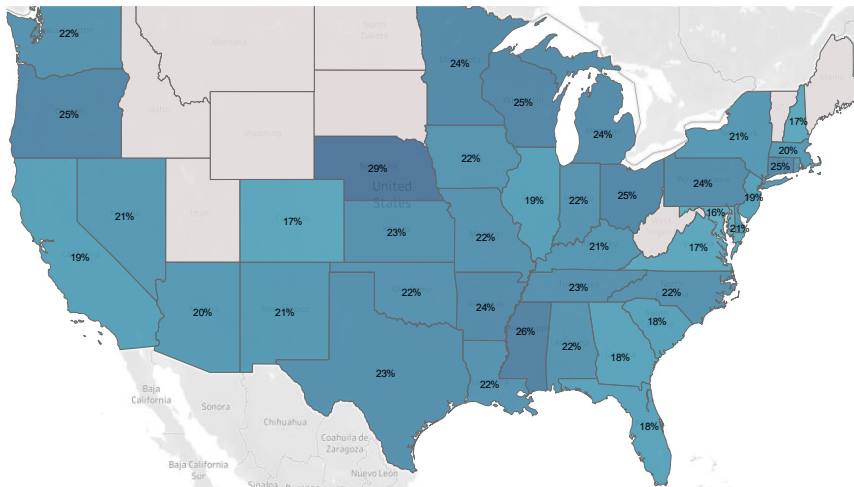
Southern states are dominated by Black and Hispanic communities...



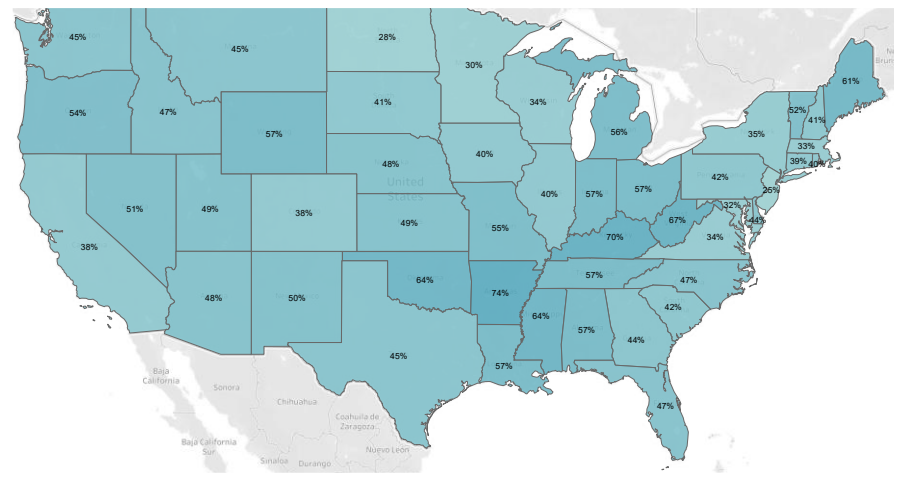
...while White populations are equally distributed across America



Across America, food insecurity is worse for Black and Hispanic populations...



...compared to White populations

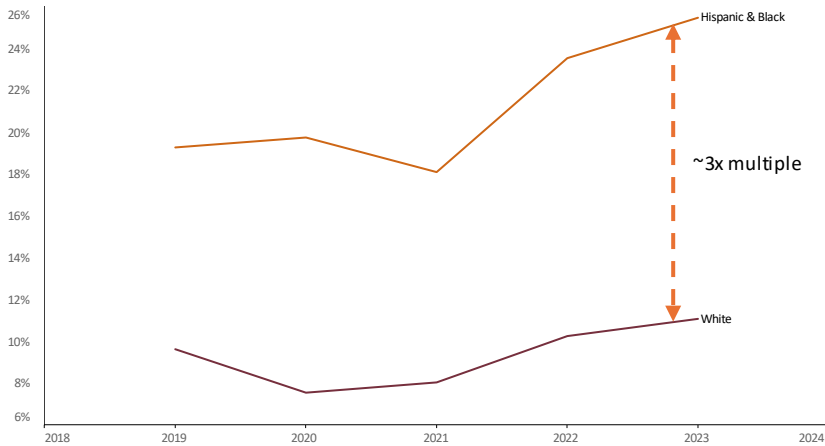


Money is the Real Problem

Hispanic and Black households experience food insecurity at 2x the rate of White households. But when normalized for income, the gap narrows sharply, revealing that the racial food divide is largely a reflection of an income divide that has barely budged in over two decades.

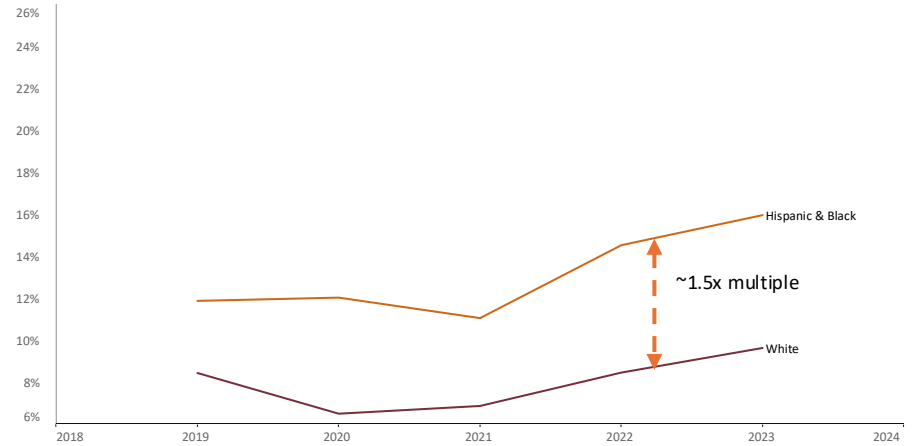
Food scarcity is ~3x worse for Hispanic and Black populations...

Rate of Food Insecurity



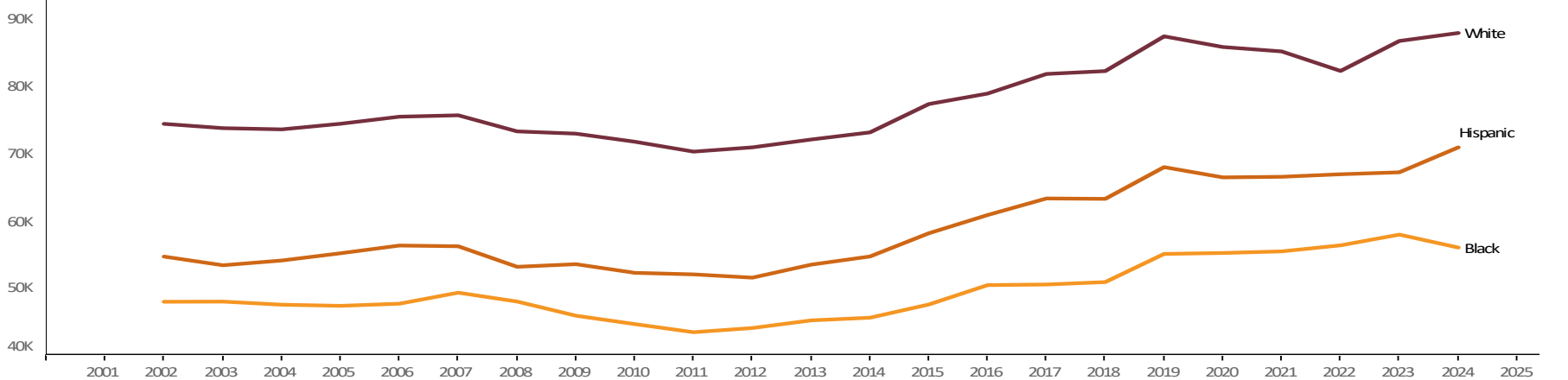
...but this gap reduces when we normalize based on median income

Rate of Food Insecurity Normalized by Median Income



Income disparity amongst racial groups has remained extremely stubborn for the past 20 years

Median Income (\$)



| Poor Communities Struggle to Buy Healthy Food

Food insecurity is not randomly distributed – it tracks poverty closely. Every increase in a county's poverty rate corresponds to a measurable rise in food insecurity, making them two faces of the same underlying problem.

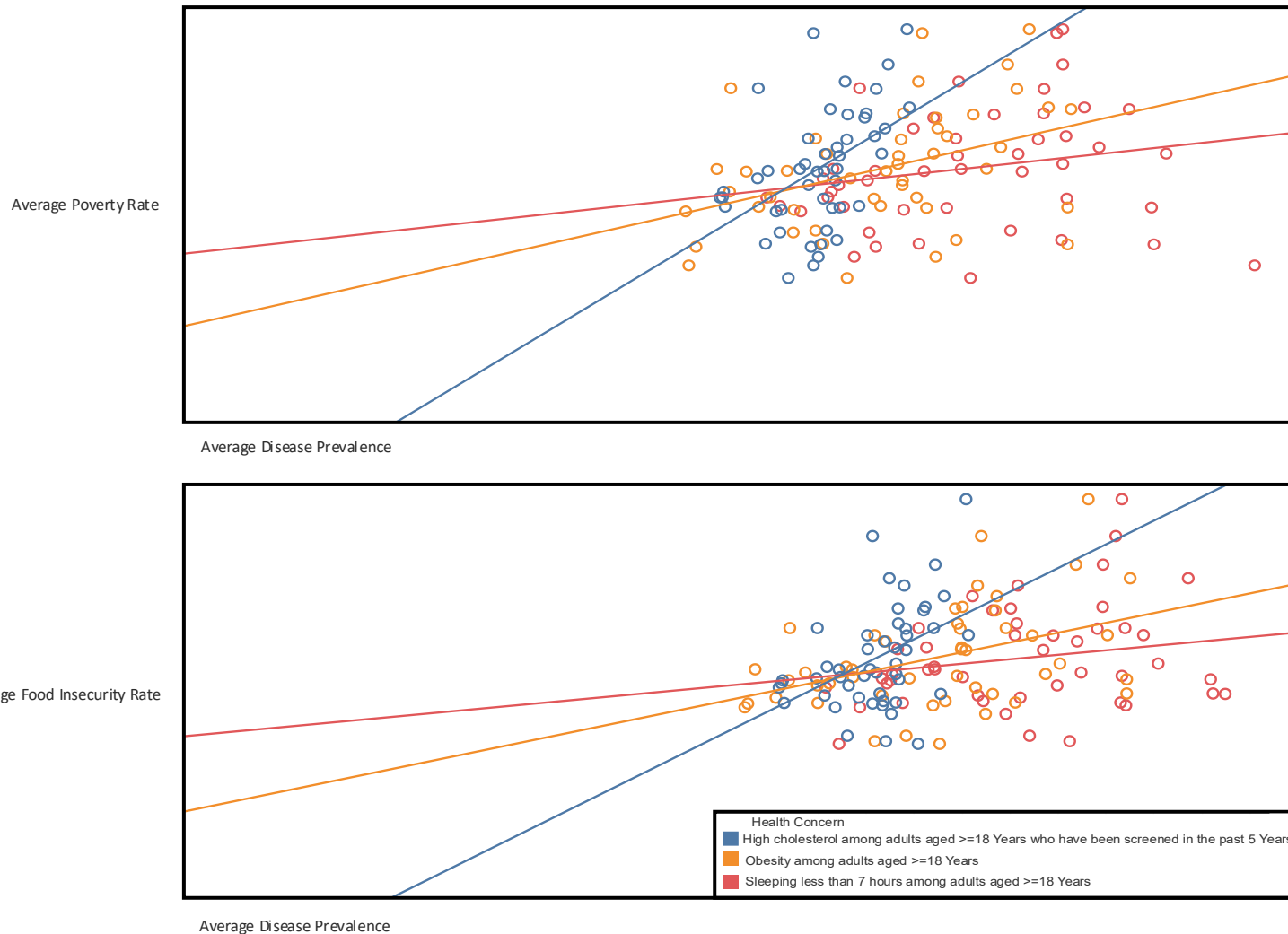
Food insecurity increases with poverty



When You Can't Eat Well, You Get Sicker

Each health concern, obesity, high cholesterol, and less sleep all follow the same pattern: counties with the highest poverty and food insecurity rates show consistently worse outcomes, suggesting these conditions do not just coexist with poor health – they shape it.

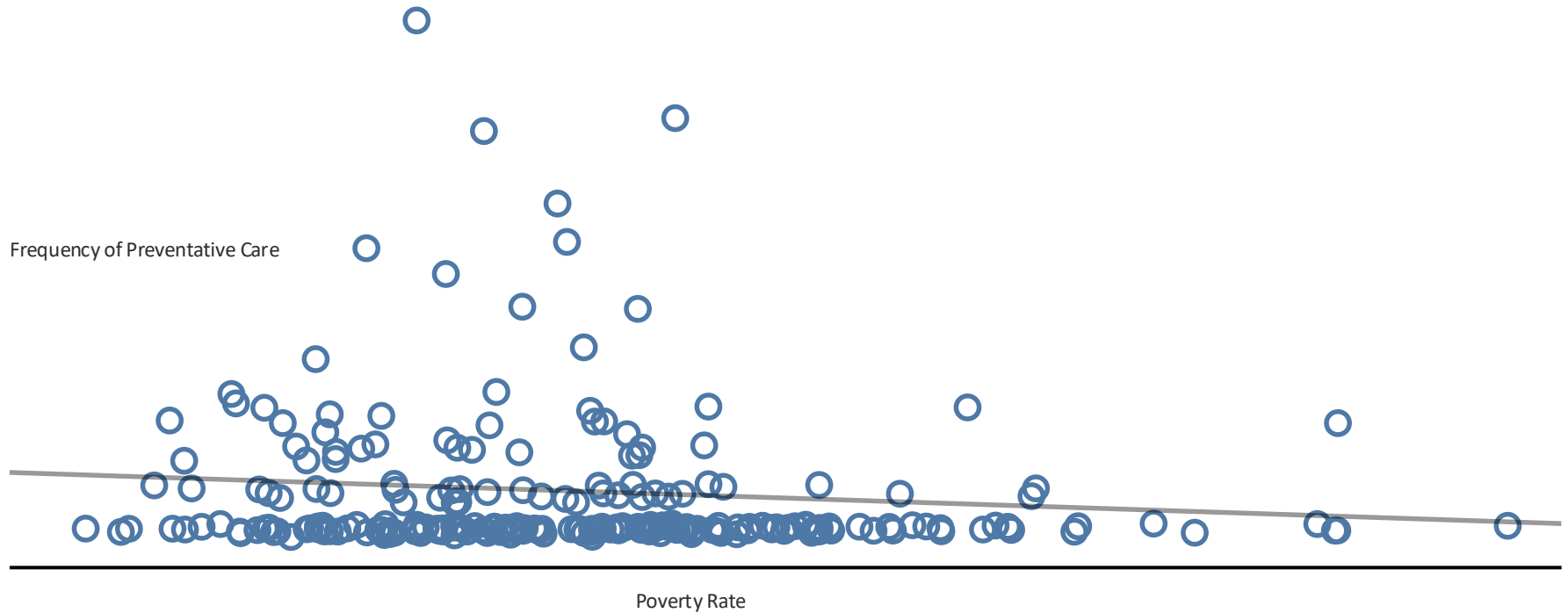
Prevalence of disease increases with higher poverty and food insecurity rates



| Access to Preventive Care Works in the Opposite Direction

Preventive care follows the opposite trajectory of poverty. Where economic hardship is highest, regular doctor visits drop off, leaving treatable conditions to progress unmonitored and untreated.

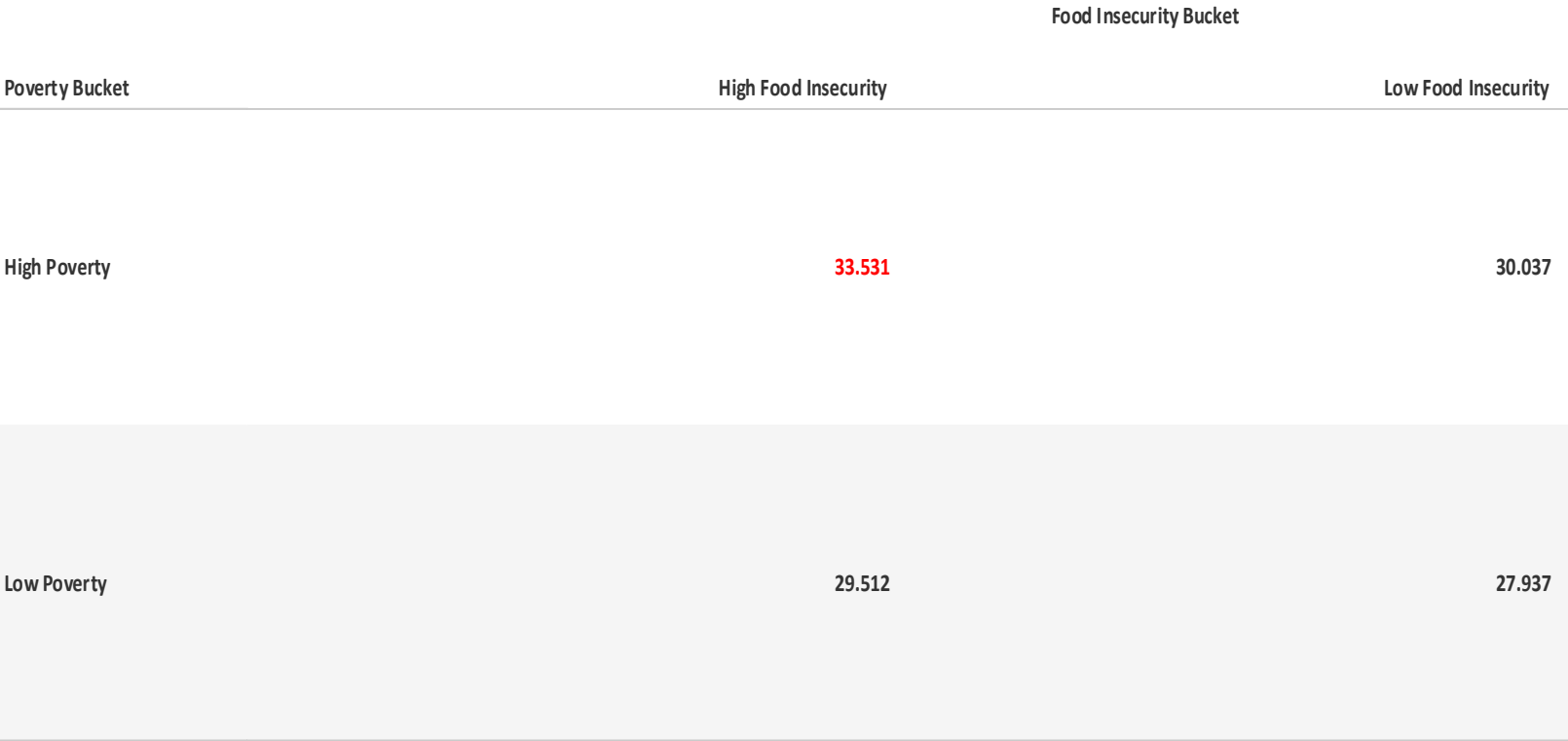
Quality of preventative care declines as poverty rate rises



Low Income and Food Access Have a Compounding Effect

Splitting counties by national medians on poverty and food insecurity reveals a clear pattern: the 'high-both' quadrant consistently shows the highest proportion of adverse health outcomes, and the gap widens beyond what either factor predicts independently.

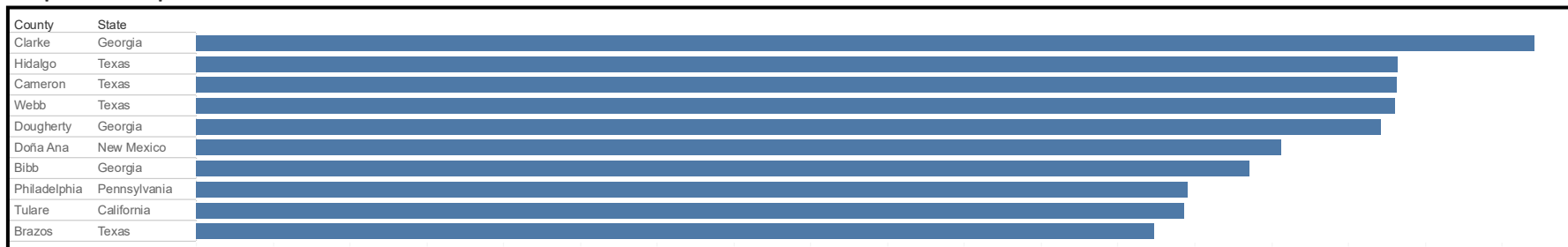
Counties Facing Both Crises See the Worst Health Outcomes



The Worst Counties Repeatedly Appear Across Multiple Metrics

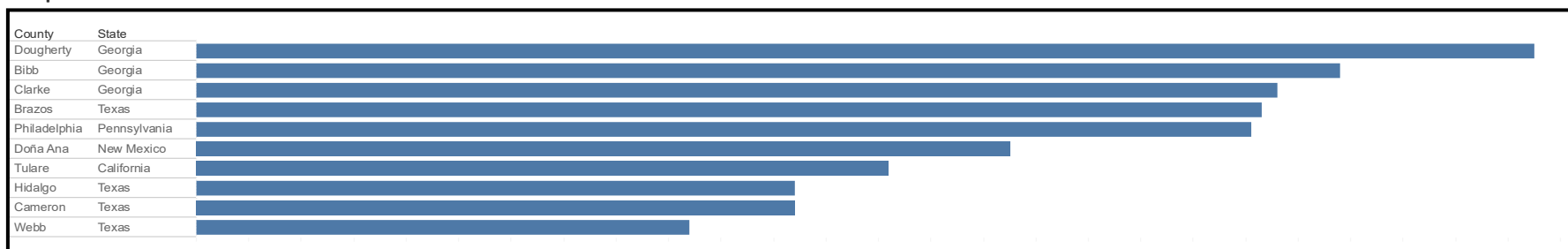
The same counties appear again and again: highest poverty, highest food insecurity and worst health outcomes. These are not separate crises happening in separate places and the overlap isn't coincidence – it's the pattern this entire analysis has been describing.

Top 10 most impoverished counties



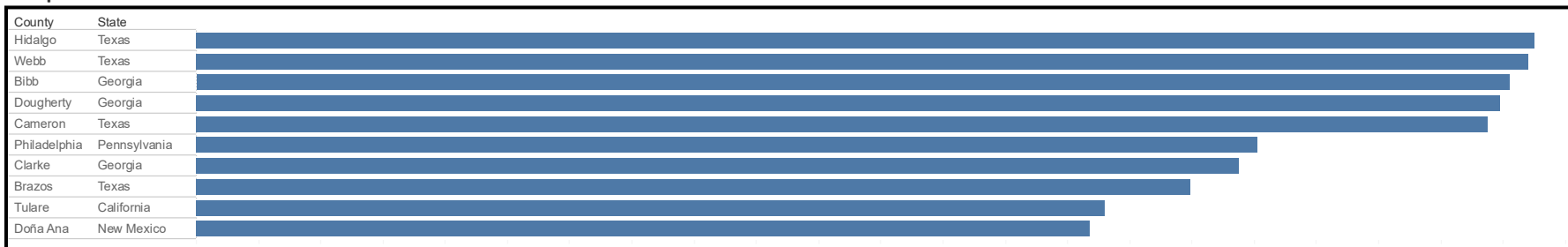
Average Poverty Rate

Top 10 most food insecure counties



Average Food Insecurity Rate

Top 10 most health concerned counties

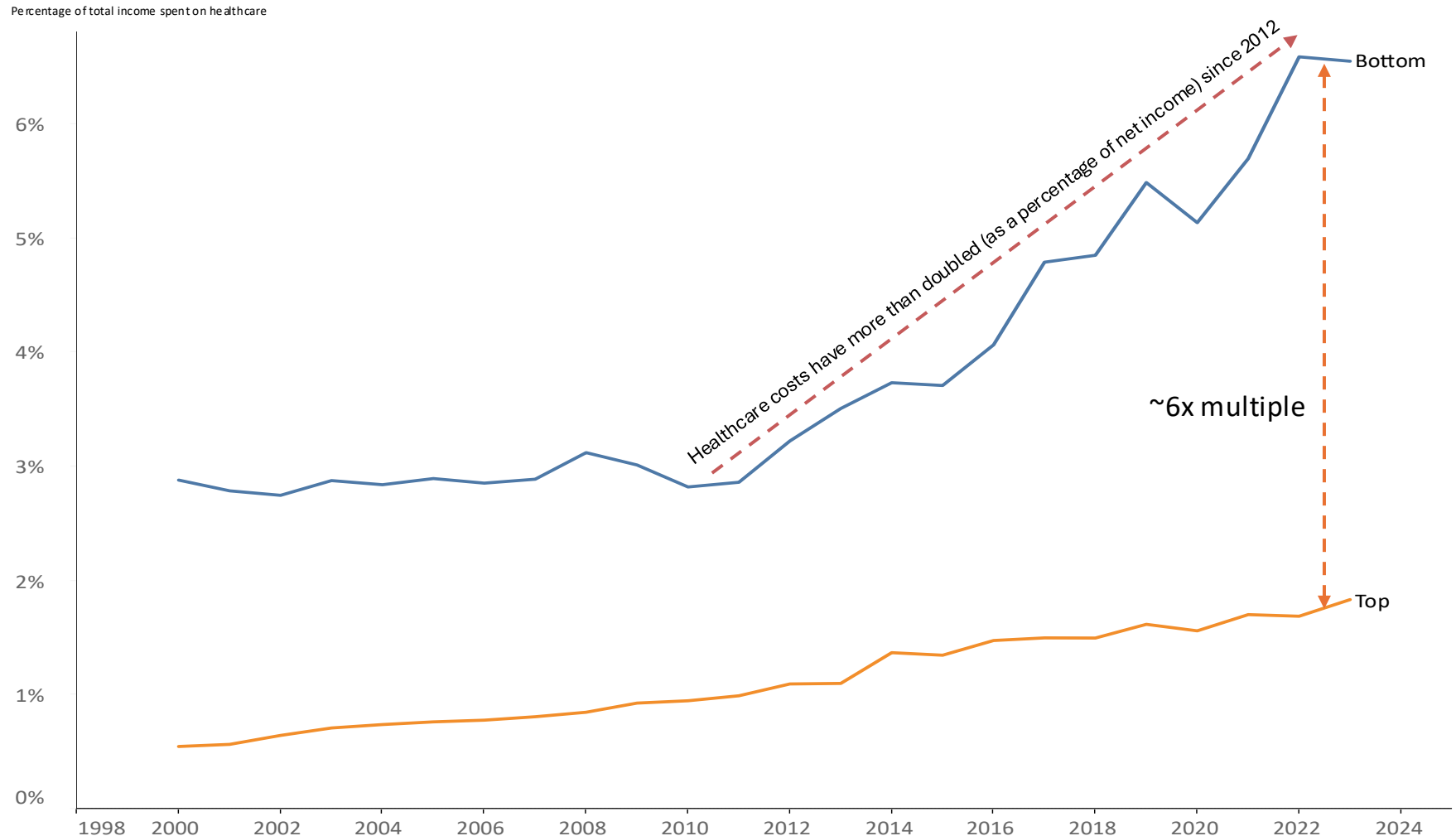


Average Disease Prevalence

| Expensive Healthcare Pushes Families Deeper into Poverty

As individuals in the lowest income brackets lose access to food and preventative care, they develop costly health crises that consume an ever-larger share of their meager income. This forces deeper cuts to nutrition and healthcare which worsens health and triggers further healthcare expenses and the vicious cycle repeats.

As a percentage of net income, healthcare costs are over 6x more for households in the bottom quintile of incomes



Key Takeaways & Conclusions

- 1 Food insecurity follows poverty and race. It is not random.
- 2 The crisis is concentrated in specific regions, especially the South
- 3 Poverty and hunger create a trap: bad nutrition leads to worse health
- 4 Medical bills hit poor families the hardest and make things worse
- 5 Focused support in the hardest-hit counties could help break this cycle

☐ **Bottom line: poor health outcomes amongst low-income households could explain why people fail to free themselves from poverty and food insecurity**